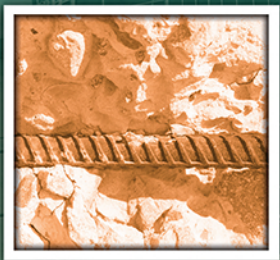


SECOND EDITION

STEEL- REINFORCED CONCRETE STRUCTURES



Assessment and Repair
of Corrosion

Mohamed Abdallah El-Reedy, PhD



CRC Press
Taylor & Francis Group

Steel-Reinforced Concrete Structures

Assessment and Repair of Corrosion

Second Edition



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*This book is dedicated to the spirits of my mother and my father,
and to my wife and my children, Maey, Hisham, and Mayar*



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Preface

Deterioration of reinforced concrete structures has been one of the major challenges facing civil engineers over the past 20 years. Concrete structures were famous for being maintenance free. Over time, however, it was found that nothing is free and that it was necessary to increase spending if one did not know how to deal with deterioration of concrete structures. The major factor causing concrete deterioration is corrosion of steel reinforcements, so it is very important to understand this issue thoroughly.

This book aims to be a handbook for reinforced concrete structure maintenance to be used as a guide for junior and senior engineers working in design, construction, repair, and maintenance. Moreover, the book has been written in a way that is easy for civil engineers to understand—without any complicated chemical reactions—and to help researchers with very strong theoretical background in the field.

The book consists of three main approaches. First, [Chapters 1 through 3](#) describe the corrosion phenomenon of steel in concrete, the effect of concrete properties on corrosion, and the precautions taken in construction to control corrosion. The second approach concerns the evaluation of concrete structures and different methods to protect steel reinforcement from corrosion. The last two chapters describe the most traditional and advanced repair techniques. Moreover, [Chapter 9](#) focuses on an advanced maintenance plan philosophy as a risk-based maintenance for reinforced concrete structures.

As the first edition was successfully distributed among different industries and is considered to be a reference for many research studies and theses, for this new edition I added the most recent research studies and development activities that were carried out in the area of corrosion of steel bars in the last 10 years and also the most recent techniques for the protection and repair of concrete structures that deteriorate due to corrosion of steel bars.

As science and hence associated technologies are updated regularly, and as the main objective of this book is to provide a practical solution with a strong theoretical background, new techniques in inspection and repair have been updated to match the current market situation.

This book is a practical guide to the art of concrete structure repair and maintenance. It includes case histories from all over the world to assist the reader in appreciating the widespread applications and range of advanced repair techniques.

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Author



Dr. Mohamed Abdallah El-Reedy's background is in structural engineering. His main area of research is the reliability of concrete and steel structures. He has provided consultation for different oil and gas industries in Egypt and to international companies, such as the International Egyptian Oil Company (IEOC) and British Petroleum (BP). Moreover, he provides different concrete and steel structure design packages for residential buildings, warehouses, and telecommunication towers. He has participated in liquefied natural gas (LNG) and

natural gas liquid (NGL) projects with international engineering firms. Currently, Dr. El-Reedy is responsible for reliability, inspection, and maintenance strategies for onshore concrete structures and offshore steel structure platforms. He has performed these tasks for hundreds of structures in the Gulf of Suez in the Red Sea.

Dr. El-Reedy has consulted with and trained executives at many organizations, including the Arabian American Oil Company, BP, Apachi, Abu Dhabi Marine Operating Company (ADMA), the Abu Dhabi National Oil Company, King Saudi's Interior Ministry, Qatar Telecom, the Egyptian General Petroleum Corporation, Saudi Arabia Basic Industries Corporation (SABIC), the Kuwait Petroleum Corporation, Qatar Petrochemical Company (QAPCO), Petroleum Development Oman (PDO), and PETRONAS Malaysian oil company. He has taught technical courses on repair and maintenance of reinforced concrete structures and steel structures worldwide, especially in the Middle and Far East.

Dr. El-Reedy has written numerous publications and presented many papers at local and international conferences sponsored by the American Society of Mechanical Engineers, the American Concrete Institute, the American Society for Testing and Materials, and the American Petroleum Institute. He has published many research papers in international technical journals and has authored eight books on total quality management, quality control and quality assurance, economic management for engineering projects, and repair and protection of reinforced concrete structures, advanced concrete materials, onshore structure design in petroleum facilities, and concrete structure reliability. He received his bachelor's degree from Cairo University in 1990, his master's degree in 1995, and his PhD from Cairo University in 2000.



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1 Introduction

The civilization of any country is measured by the advances in the techniques used in the construction of its buildings. Concrete is the main element used in construction materials and the quality of concrete being used for construction in any country is an indication of that country's progress in engineering thinking. Ancient Egypt used concrete in its buildings and temples; Egyptians used crushed stone as an aggregate and clay as an adhesive. The Greeks used concrete in their buildings and called it *Santorin Tofa* (El-Arian and Atta 1974); the Romans used a material called *pozzolana* that resembled concrete. After that, the formula for concrete was lost for many centuries. In the eighteenth century, a number of famous scientists worked to create a formula for concrete:

- John Smeaton used concrete to construct the Eddystone Lighthouse.
- Joseph Parker researched on stone and its uses in concrete.
- Edgar researched using cement made from limestone and clay.
- Louis Vicat also tried to develop cement from limestone and clay.
- Joseph Aspdin completed his research by developing portland cement.

At the end of the nineteenth century and in the beginning of the twentieth, the architectural shapes of buildings underwent major changes as architectural engineers and builders changed their points of view. Previously, ideas from the European Renaissance of using columns and arches to move toward the functional use of buildings have been implemented. Concrete has been found to be the best choice for functional use and to achieve architectural dreams using an economic approach.

Currently, reinforced concrete is considered to be the most popular and important material in the construction industry. It is used for buildings and for different types of civil engineering projects, such as tunnels, bridges, airports, and drainage and hydraulic projects. Research has been focused on increasing the performance of concrete to match the wide varieties of applications.

Reinforced concrete is considered to be inexpensive compared to other building materials; thus, it has been used for high-rise buildings for megaprojects and also for smaller projects, such as one-story buildings; all these projects are handled by contractors and engineers with different capabilities. Therefore, sometimes, the reinforced concrete does not perform correctly according to specifications; in some cases, there is a difference between the quality required by the standard or code and that found on the site. These errors usually arise due to a lack of worker, foreman, or engineer competence. Other construction errors arise due to a lack of quality-control procedures, materials received at the site that are incompatible with the standard, improper selection of materials, or a concrete mix that does not fit the surrounding environmental conditions.

The main challenge that faces the engineer working in the concrete industry today is the corrosion of steel reinforcement bars. When concrete was first used, concrete structures were traditionally treated as structures that were maintenance free during their lifetime. For many years, it has been obvious that this viewpoint is false, although concrete is indeed a durable material. Our dreams have broken on the rock of truth as we have found that a number of concrete structures have undergone major deterioration because of different factors and that 80% of the deterioration of reinforced concrete structures is due to the corrosion of the steel-reinforced bars. Based on this fact, much research and many development techniques have focused on providing materials and new methodology to protect steel reinforcement bars from corrosion.

Because severe climatic conditions are the major contributing factor of corrosion in reinforcement steel bars, all specifications and codes recommend some precautions to be taken in concrete mix design, material selection, concrete cover thickness, and others in order to maintain the durability of concrete during its lifetime and to prevent corrosion of the steel bars based on different environmental conditions. Corrosion of steel is the enemy of any country's investment in real estate; some structures are destroyed completely due to corrosion and depleted steel bars have a direct impact on the safety of a structure.

Billions of dollars have been spent on the corrosion problem worldwide. For example, the United States has spent billions of dollars to repair bridge decks. In North Africa, near the coast, and in the Middle East in the Arabian Gulf area, some buildings have been completely destroyed due to deterioration of the structures as a result of corrosion in the steel reinforcement bars.

It is important to know the causes of corrosion, the concrete parameters, and the environmental conditions surrounding the structure that trigger corrosion and affect the corrosion rate. For instance, it is acceptable in European and Middle Eastern countries to use seawater in the concrete mix. Also, in 1960, calcium chloride was used as an additive to assist in concrete-setting acceleration, and seawater was used in concrete mix until 1970. After that, much damage to the concrete structure was found over a 20-year period—especially in Middle Eastern countries where, because potable water is very expensive, they used seawater in most of the concrete mixes for different structures. This period had a very bad impact from an economic point of view because the investment for the construction of concrete structures was destroyed. From the above facts, the risk of corrosion and its impact on the national income of any country becomes clear, and therefore, an understanding of corrosion characteristics and causes of corrosion, which are discussed, respectively, in [Chapters 2](#) and [3](#), becomes important.

Usually, corrosion of steel bars is indicated by brown spots on the concrete surface; in some cases, cracks parallel to the steel bars will be seen and, in the worst case, the steel bars can be seen directly because the concrete cover has failed. When corrosion occurs, its major impact on structure safety will be the danger of the falling of concrete cover, as is the case when the structure loses its strength due to reduction in the concrete cross-section dimension and also reduction in the cross section of the steel bars due to corrosion. The spalling of concrete cover happens because the corroded steel bars increase in volume and exert high stress on the concrete cover; this causes cracking and subsequent falling of the cover.

Usually, a structural engineer who has a knowledge of the theoretical and practical methods of performing structural assessment is requested to assess the existing building. This will be discussed in [Chapter 4](#). All the costs for repair and reconstruction have a higher impact on a country's economy in general because the money paid for repair and reconstruction could have gone into other investments.

[Chapter 5](#) discusses different codes and standards that deal with corrosion and provides precautions to be taken in design and construction to reduce the risk of corrosion in a structure. Corrosion control is illustrated in [Chapter 6](#). [Chapter 7](#) provides all the steps for repair based on standards, traditional and advanced techniques of repair, and ways to choose the best method of repair. There are many research and development techniques to protect steel reinforcement bars; all the practical methods available on the market will be illustrated in [Chapter 8](#).

During the last 20 years, some researchers have studied the reliability of reinforced concrete structures and have provided different techniques, such as qualitative and quantitative risk assessments. In addition, the maintenance strategy and plan have been reviewed in different studies, based on risk-based maintenance. Our aim is to use an optimal methodology that reduces maintenance cost to a minimum, and also maintains the structure within a reliability limit that enables functional use to user satisfaction. Methods that will assist in implementing the maintenance plan from an economic point of view, as well as the economic method of choice among steel protection alternatives, will be discussed in the last chapter.

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